D'Tee & Gee Kitchen Website Implementation Guide

This guide provides step-by-step instructions on how to set up and run the D'Tee & Gee Kitchen e-commerce website on your local machine using VS Code. Follow these instructions carefully to ensure a smooth setup process.

1. Prerequisites

Before you begin, ensure you have the following software installed on your system:

- **Visual Studio Code (VS Code):** A powerful and free code editor. Download from https://code.visualstudio.com/
- **Python 3.8+:** For the Flask backend. Download from https://www.python.org/downloads/
- **Node.js and npm (or Yarn):** For managing frontend dependencies (though for this project, we're using pure HTML/CSS/JS, npm is useful for general web development tools). Download from https://nodejs.org/en/download/
- **Git:** For version control (optional, but good practice for managing code). Download from https://git-scm.com/downloads

2. Get the Source Code

First, you need to get the website's source code onto your machine. You can do this by either cloning a Git repository (if you have one) or by manually copying the provided source code files.

Option A: Using Git (Recommended)

If you have the project in a Git repository, open your terminal or VS Code's integrated terminal (`Ctrl+\`` or Cmd+``) and run:

```
git clone <your-repository-url>
cd dtee-gee-kitchen
```

Replace `<your-repository-url>` with the actual URL of your Git repository.

Option B: Manual Copy

If you received the source code as a zipped file or individual files, create a new folder on your computer (e.g., `dtee-gee-kitchen`) and extract/copy all the provided files into this folder.

3. Open the Project in VS Code

1. Open VS Code.
2. Go to `File > Open Folder...` (or `Cmd+Shift+O` on macOS).
3. Navigate to the `dtee-gee-kitchen` folder you created/cloned and click `Open`.

4. Backend Setup (Python Flask)

The backend is built with Python Flask and uses SQLite as its database. We'll set up a virtual environment to manage dependencies.

1. **Open a new terminal in VS Code:** Go to `Terminal > New Terminal`.
2. **Navigate to the backend directory:** `bash cd backend`
3. **Create a Python virtual environment:** `bash python3 -m venv venv`
   - If `python3` doesn't work, try `python`.
4. **Activate the virtual environment:**
   - **On macOS/Linux:** `bash source venv/bin/activate`
   - **On Windows (Command Prompt):** `bash venv\Scripts\activate.bat`
   - **On Windows (PowerShell):** `powershell .\venv\Scripts\Activate.ps1` You should see `(venv)` at the beginning of your terminal prompt, indicating the virtual environment is active.
5. **Install Python dependencies:** `bash pip install -r requirements.txt`
   - If `requirements.txt` is not provided, you will need to install the necessary libraries manually: `bash pip install Flask Flask-SQLAlchemy Flask-CORS Werkzeug`
6. **Initialize the database and seed data:** The project uses a SQLite database (`app.db`) which will be created automatically. The `seed_data.py` script populates it with initial categories and products. `bash python seed_data.py`
   - If you encounter an error about the database already existing, you might need to delete the old `app.db` file first (located in `backend/src/database/`). `bash rm src/database/app.db # On macOS/Linux del src\database\app.db # On Windows python seed_data.py`

7. **Run the Flask backend server:** `bash python src/main.py` You should see output indicating the Flask app is running, typically on `http://127.0.0.1:5000` or `http://localhost:5000`.

   Leave this terminal open and running. The backend server needs to be active for the frontend to work.

5. Frontend Setup (HTML, CSS, JavaScript)

The frontend is a static website served by a simple Python HTTP server. This allows for easy local testing.

1. **Open a new terminal in VS Code:** Go to `Terminal > New Terminal` (do not close the backend terminal).
2. **Navigate to the project root directory:** `bash cd /home/ubuntu/dtee-gee-kitchen` (If you are already in the root, you don't need to `cd`.)

3. **Run the frontend server:** `bash python serve_frontend.py 3000` This will start a simple HTTP server serving the `frontend` directory on port `3000`.

   Leave this terminal open and running.

6. Access the Website

Once both the backend and frontend servers are running, open your web browser and navigate to:

http://localhost:3000

You should see the D'Tee & Gee Kitchen website loading. If you see a loading screen, it might take a moment for the content to appear as it fetches data from the backend.

7. Important Notes & Troubleshooting

- **Backend Port:** The backend runs on port `5000`. The frontend will communicate with it on this port. If you change the backend port, you'll need to update `frontend/js/api.js` to reflect the new port.
- **Frontend Port:** The frontend runs on port `3000`. If this port is in use, the `serve_frontend.py` script might fail. You can try a different port by running `python serve_frontend.py <new_port_number>`.
- **CORS Issues:** If you encounter